

Cryptanalysis

- Study of breaking encryption
- Types: brute force, frequency analysis
- Goal: find key or original message

Modern Cryptography

- Uses mathematics and algorithms
- More secure than classical methods
- Used in banking, apps, internet

Symmetric Cryptography

- One key for encryption & decryption
- Fast and efficient
- Example: AES

Asymmetric Cryptography

- Two keys: public and private
- More secure for communication
- Example: RSA

Symmetric vs Asymmetric

- Symmetric: one key, fast
- Asymmetric: two keys, secure
- Used together in real systems

AES Introduction

- Advanced Encryption Standard
- Symmetric algorithm
- Used in WiFi, banking, file security

RSA Introduction

- Asymmetric algorithm
- Uses public & private keys
- Based on large prime numbers